Problem Statement No.11 – AI Lab Manual & Experiment Generator

The Challenge – An AI Lab Manual & Experiment Generator, powered by RAG (Retrieval-Augmented Generation), assists educators and students by generating step-by-step lab manuals, virtual experiment simulations, and evaluation rubrics tailored to their syllabus and available equipment.

It retrieves relevant curriculum data, lab protocols, and assessment guidelines from academic databases and institutional repositories.

Users can input their course code or syllabus topics and receive ready-to-use manuals, safety instructions, and grading criteria.

The agent ensures consistency, accuracy, and alignment with educational standards, reducing the manual effort of lab preparation.

This AI-driven assistant enhances hands-on learning, supports remote labs, and modernizes STEM education delivery.

Technology – Use of IBM Cloud Lite services / IBM Granite is mandatory.